Unit 2 – Exploring Derivatives

THE CONSTANT RULE

If $f(x)=c$, c is a constant, then $f'(x)=0$.

Proof:

Example: Determine the derivative for each of the following.

a. $f(x)=0$ b. $f(x)=\sqrt{7}$ c. $y=-\frac{1}{2}$ d. $y=-4.67$

THE POWER RULE

If $f(x)=x^n$, where $n\in\mathbb{R}$, then $f'(x)=nx^{n-1}$.

Proof:

Let $f(x)=x^n$

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$=\lim_{h\to 0}\frac{(x+h)^n-x^n}{h}$$

$$=\lim_{h\to 0}\frac{\cancel{x^n}+nx^{n-1}(h)+\frac{n(n-1)}{2}x^{n-2}(h^2)...+(h)^n\cancel{-x^n}}{h}$$

Factor out the "h"

$$=\lim_{h\to 0}\frac{\cancel{h}[nx^{n-1}+\frac{n(n-1)}{2}x^{n-2}(h)...+(h^{n-1})]}{\cancel{h}}$$

Let "h" go to zero

$$=nx^{n-1}$$

Example: Determine the derivative for each of the following.

a. $f(x)=x$

b. $f(x)=x^5$

c. $y=\sqrt{x}$

d. $y=x^{-4}$

e. $f(x)=\frac{1}{\sqrt[3]{x}}$

f. $f(x)=\sqrt[6]{x}$

THE CONSTANT MULTIPLE RULE

If $f(x) = c \cdot g(x)$, c is a constant, then $f'(x) = c \cdot g'(x)$.

Example: Determine the derivative for each of the following.

a. $f(x) = 4x^3$

b. $f(x) = -16x^{-4}$

c. $y = -2x^{\frac{3}{2}}$

d. $y = \sqrt{6}x^{-\frac{3}{5}}$

e. $y = 3x^{\sqrt{5}}$

f. $y = \pi x^{\pi}$

SUM & DIFFERENCE RULES

If both $p(x)$ and $q(x)$ are differentiable and $f(x) = p(x) \pm q(x)$, then $f'(x) = p'(x) \pm q'(x)$.

Example: Determine the derivative for each of the following.

a. $f(x) = \sqrt{\dfrac{2}{x}} + \sqrt{\dfrac{x}{3}}$

b. $y = -\dfrac{\sqrt{2}}{2} x^{-\frac{3}{2}} + \dfrac{x^{-\frac{1}{2}}}{2\sqrt{3}}$

c. $y = x^{\pi} - \pi x^{3}$

d. $f(x) = 7x^{4} - \sqrt{x} + 3x^{\frac{3}{2}} - 401$

Example: Determine the equation of the tangent to the graph of $y = x^3 + 2x^2 - 4x + 1$ at $x = 4$.

Example: Determine the values(s) of x so that the tangent to $f(x) = \frac{3}{\sqrt[3]{x}}$ is parallel to the line $x + 16y + 3 = 0$.

Additional Questions

1. Determine the derivative of each of the following.

a) $f(x) = -\frac{3}{4}\sqrt[4]{x^5} - \frac{4}{3\sqrt{x^3}} + \pi^2 x^3 - \frac{7}{3}$

b) $g(x) = 4\sqrt[4]{x^3}\left(\pi\sqrt[5]{x} - 2^3\pi\right)$

c) $g(x) = \frac{3x^4 + 2\pi x^3 - 5\sqrt[3]{x}}{4x^2}$